Ahsan Bilal

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RESEARCH INTERESTS

Deep Learning Optimization and LLM Inference: Optimization theory for deep learning and reasoning-oriented LLMs. LLM post-training and inference methods including preference optimization, active learning, alignment, and test-time compute scaling for autoregressive and diffusion language models.

Agentic and Adaptive AI Systems: Design of evolving multi-agent systems with adaptive behaviors, dynamic reputation modeling, and social graph–based coordination.

Reinforcement Learning for Reasoning and Decision Systems: RL methods for reasoning-oriented LLMs, process reward models (PRMs), and agentic test-time compute allocation, with applications in complex non-stationary environments and ML-driven wireless communication systems.

EDUCATION

University of Oklahoma, Oklahoma, USA — 2024–Present
MS in Computer Science (Deep Learning, Reinforcement Learning, & LLMs)

- CGPA: 3.86/4.0
- Advisor: Dr. Dean Hougen: Research on deep learning optimization, robust LLM reasoning, and RL-based agentic systems.

National University of Sciences & Technology (NUST-SEECS), Islamabad, Pakistan — Aug. 2020 – June 2024
Bachelor of Engineering in Electrical Engineering

- CGPA: 3.66/4.0 | Specialization GPA: 4.0/4.0 | Merit Scholarship 2020
- Senior Design Thesis: Person identification using gait with fused graph and 3D-convolutional architectures (Presentation)

RESEARCH EXPERIENCE

Graduate Researcher — REAL Lab, University of Oklahoma
Advisor: Dr. Dean Hougen — *Aug 2024 – Present*

- Research on structured generative models and RL-driven reasoning for LLMs (targeting ICML'26), including discrete diffusion models and agentic RL controllers for test-time compute optimization.
- Contributed to continual-learning and diffusion-based wireless channel prediction models (two ICASSP'26 submissions).

Research Collaborator — MLCN Lab, Stanford University
Supervisor: Prof. John M. Cioffi | Collaborator: M. A. Mohsin — *Aug 2024 – Present*

- Developed diffusion-based wireless estimation, neural fields (nGRF), continual learning under distribution shift, and RAG pipelines for multimodal wireless systems.
- Publications: ICLR'26 (under review), NeurIPS'25, ICML'25, AAAI'25, ICC'25 (Best Paper).

Undergraduate Researcher — Optimal ML Lab, NUST
Supervisor: Dr. Ahmad Salman — *Jan 2023 – June 2024*

- Built robust CV/biometric models: boosted-attention ViT for shadow removal and fused GCN + 3D-CNN for gait recognition.
- Papers under review: Shadow Removal with Boosted Attention, Gait ID using Fused Graph + 3D-CNN.

SELECTED PUBLICATIONS

Google Scholar

[PC1] Internet of Evolving Agents NeurIPS'26 (In Progress)
Z. Ali*, M.T. Shah*, M.A. Mohsin*, M. Umer, A. Bilal, E. Fox

[PC2] Sycophancy Disentanglement in LLMs via Reward Decomposition NeurIPS'26 (In Progress)
M.A. Mohsin*, A. Bilal*, M. Umer, E. Fox

[PC3] S3: Stratified Scaling Search for Test-Time in Diffusion Language Models CoLM'26 (In Progress)
A. Bilal*, M.A. Mohsin*, M. Umer, D.F. Hougen

[PC4] What If We Allocate Test-Time Compute Adaptively? ICML'26 (Submitted)
A. Bilal*, A. Mohsin*, M. Umer, A. Subhan, H. Rizwan, A. Mohsin, D.F. Hougen

[PC5] Continuous-Utility Direct Preference Optimization ICML'26 (Submitted)
M.A. Mohsin*, M. Umer*, A. Bilal, Z. He, M.U. Rafique, A. Aali, M.A. Jamshed, J.M. Cioffi, E. Fox

[PC6] ITDPDM: Information-Theoretic Discrete Poisson Diffusion Model NeurIPS'25
S. Bhattacharya, A.R. Gorle, A. Bilal, C. Ding, A.K.S. Yadav, T. Weissman

[PC7] On the Fundamental Limits of LLMs at Scale TMLR'26 (Submitted)
A. Mohsin, A. Bilal, W. Zhao, M. Umer, Researchers from DeepMind and Meta

[PC8] Neural Gaussian Radio Fields for Channel Estimation KDD'26 (Submitted)
M. Umer*, M.A. Mohsin*, A. Bilal*, J.M. Cioffi

[PC9] Channel Prediction Under Network Distribution Shift Using Continual Learning-Based Loss Regularization ICASSP'26
M.A. Mohsin, M. Umer, A. Bilal, M.I. Qadir, M.A. Jamshed, D.F. Hougen, J.M. Cioffi

[PC10] Conditional Prior-Based Non-Stationary Channel Estimation Using Accelerated Diffusion Models ICASSP'26
M.A. Mohsin*, A. Bilal*, M. Umer, A. Ali, M.A. Jamshed, D.F. Hougen, J.M. Cioffi

[PC11] Transformer-Based Sparse CSI Estimation for Non-Stationary Channels ICC'26
M.A. Mohsin, M. Umer, A. Bilal, H. Rizwan, S. Bhattacharya, M.A. Jamshed, J.M. Cioffi

[PC12] Continual Learning for Wireless Channel Prediction ICML'25
M.A. Mohsin, M. Umer, A. Bilal, M.A. Jamshed, J.M. Cioffi

[PC13] Task Aware Distributed Source Coding for Correlated Audio Signals Using Perceptual Loss AAAI'25
S. Bhattacharya, M.A. Mohsin, A. Bilal, J.M. Cioffi

[PC14] Retrieval Augmented Generation with Multi-Modal LLM Framework for Wireless Environments ICC'25
M.A. Mohsin, A. Bilal, S. Bhattacharya, J.M. Cioffi

* indicates equal contribution.

[PC15] HDRL for Spectrum Resource Optimization in Integrated Terrestrial and Non-Terrestrial Networks AAAI'25
M.A. Mohsin, H. Rizwan, M. Umer, S. Bhattacharya, A. Bilal, J.M. Cioffi

[PC16] Abstract – LLM for Explainable AI IEEE DSAA'24
A. Bilal, B. Lin

[PC17] Meta-Thinking in LLMs via Multi-Agent Reinforcement Learning: A Survey IEEE TAI (Submitted)
A. Bilal, M.A. Mohsin, M. Umer, M.A.K. Bangash, M.A. Jamshed

[PC18] On Shadow Removal With Boosted Attention in a Vision Transformer Springer ML (Submitted)
A. Bilal, A. Salman, K. Khurshid, D.F. Hougen

[PC19] Person Identification using Gait with Fused Graph and 3D-Convolutional Architectures ACM TAIS (Submitted)
A. Bilal, A. Salman, K. Khurshid

INDUSTRY EXPERIENCE

Machine Learning Engineer Islamabad, Pakistan
Cowlar Design Studio (Y Combinator 21) — Based in USA *Feb 2024 – Aug 2024*

- Developed Action Recognition system for Smart Carts with 95% accuracy; built dual inference deployment for edge devices and Nvidia cluster.
- Automated fiber cable alignment using computer vision and ML with 96% success, improving precision to 5 micrometers and scaling production 40x.

UI Developer Dubai, UAE (Remote)
SJCurve *March 2023 – Aug 2024*

- Created responsive web applications using WordPress and React.js, customized themes, and integrated frontend with backend.

UX/UI Designer Wah Cantt, Pakistan (Remote)
Meraki-IT *Nov 2022 – Mar 2023*

- Led UI/UX design for multiple projects, analyzing problem statements and designing intuitive interfaces for technology sector.

TEACHING EXPERIENCE

Teaching Assistant University of Oklahoma
CS-1313: Programming for Non-majors in C *Fall 2024 – Present*

- Worked with Dr. Neeman to design weekly lab assignments, grade with clear rubrics, and lead help sessions supporting students with C programming.

REVIEWER AND TALKS

Conference Reviewer: ICASSP, PAKDD.

Journal Reviewer: TMLR, IEEE WCM, Springer MT&A, IP&M, Aquaculture Int., IJIM, IEEE Access, ICES, ISFI.

Talks: Gave a talk on "AI in Healthcare" at Norman Regional Hospital under Dr. Lubna Mirza.

HONORS AND ACHIEVEMENTS

Honorary Certificate of Appreciation: IEEE Communications Society Student Competition 2025 for "Democratizing 6G: AI-Native Wireless Digital Twin for Global Digital Equity and Sustainability."

Best Student Presentation Runner-up Award: IEEE DSAA'24 Student Forum.

Graduate Fellowship: Awarded Gallogly College of Engineering Graduate Fellowship 2025.

Best Paper Award: ICC Workshop 2025 in Montreal.

Student Travel Grant: IEEE DSAA 2024 in San Diego.

Best Adjudged Industrial Project Award: Final Year Project received 1st place at NUST Open House 2024.

UGRIP Selection: Selected as Undergraduate Research Intern for First Cohort by MBZUAI.

IEEE Recognition: Selected as Emerging Young Researcher in IEEE Islamabad Section.

Prime Minister Laptop Scheme: Winner of scholarship program.

CERTIFICATIONS

Google UX Professional Certificate

Deep Learning Specialization Certificate

The Advanced Communication Skills Course

SKILLS

Programming: Python, C/C++, Embedded C, MATLAB.

Deep Learning: PyTorch, TensorFlow (Keras), OpenCV, Docker, Mlflow, EC2 Instance.

Web Development: Javascript (React.js), HTML/CSS, Next.js.

Tools: VS Code, Git, AutoCad, Figma, PyCharm, Raspberry Pi OS, NginX.

Design: Figma, AdobeXD, Adobe Illustrator, Adobe Photoshop, Sketch, WordPress Theme Design.